

- Description

This module manages an elevator overload detection system, ensuring that:

1. The system continuously monitors the total weight inside the elevator (in kg).
2. If the total weight exceeds the maximum load capacity, an overload alarm is triggered, preventing the elevator from moving.
3. If the weight is within the safe limit, the elevator operates normally..

This system ensures passenger safety by preventing elevator operation under overloaded conditions.

- Specification

module SmartElevatorOverloadDetection

var currentLoad, maxLoadCapacity: real

process updateLoad(newLoad: real) returns (currentLoad: real)

functional scenario form:

currentLoad > maxLoadCapacity && output "ALERT: Overload detected (" + currentLoad + " kg). Elevator cannot operate.", Overload alarm triggered, elevator operation blocked.

||

currentLoad <= maxLoadCapacity && output "Current load: " + currentLoad + " kg. Elevator operating normally.", Elevator operates normally.

end process

end module

- Java Code

```
package code;
public class c96_SmartElevator {
    private double maxLoadCapacity;
    private double currentLoad;
    public c96_SmartElevator(double maxLoadCapacity) {
        this.maxLoadCapacity = maxLoadCapacity;
        this.currentLoad = 0.0;
    }
    public void updateLoad(double newLoad) {
        currentLoad = newLoad;
        System.out.printf("Current Load: %.2f kg%n", currentLoad);

        if (currentLoad > maxLoadCapacity) {
            System.out.println("ALERT: Overload detected! Elevator cannot operate.");
        } else {
            System.out.println("Elevator operating normally.");
        }
    }
}
```

```

    public double getCurrentLoad() {
        return currentLoad;
    }
    public static void main(String[] args) {
        c96_SmartElevator elevator = new c96_SmartElevator(800.0);
        // testcase-VT:
//        elevator.updateLoad(750.0);
//        elevator.updateLoad(850.0);
//        elevator.updateLoad(800.0);

        // testcase-FT:
//        elevator.updateLoad(-1.7836486571137695E308);
//        elevator.updateLoad(-1.2359056723264467E308);
//        elevator.updateLoad(-1.193771596563137E308);

        // testcase-Z3:
//        elevator.updateLoad(750.0);
//        elevator.updateLoad(850.0);
//        elevator.updateLoad(800.0);

        // testcase-UVT:
        elevator.updateLoad(750.0);
        elevator.updateLoad(850.0);
        elevator.updateLoad(800.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c96_SmartElevator.java	100.0 %	2	0	2
✓ c96_SmartElevator	100.0 %	2	0	2
main(String[])		0	0	0
c96_SmartElevator(double)		0	0	0
updateLoad(double)	100.0 %	2	0	2

3. Vibration testing instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.2 %	53	21,686	21,739
✓ src	0.2 %	53	21,686	21,739
✓ code	0.2 %	53	21,686	21,739
✓ c96_SmartElevator.java	94.6 %	53	3	56
✓ c96_SmartElevator	94.6 %	53	3	56
main(String[])	100.0 %	15	0	15
c96_SmartElevator(double)	100.0 %	9	0	9
updateLoad(double)	100.0 %	29	0	29

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.1 %	1	841	842
▼ src	0.1 %	1	841	842
▼ code	0.1 %	1	841	842
▼ c96_SmartElevator.java	50.0 %	1	1	2
▼ c96_SmartElevator	50.0 %	1	1	2
● updateLoad(double)	50.0 %	1	1	2
● c96_SmartElevator(double)		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 87.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	49	21,690	21,739
▼ src	0.2 %	49	21,690	21,739
▼ code	0.2 %	49	21,690	21,739
▼ c96_SmartElevator.java	87.5 %	49	7	56
▼ c96_SmartElevator	87.5 %	49	7	56
● updateLoad(double)	86.2 %	25	4	29
● c96_SmartElevator(double)	100.0 %	9	0	9
● main(String[])	100.0 %	15	0	15

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c96_SmartElevator.java	100.0 %	2	0	2
▼ c96_SmartElevator	100.0 %	2	0	2
● main(String[])		0	0	0
● c96_SmartElevator(double)		0	0	0
● updateLoad(double)	100.0 %	2	0	2

7. Z3 instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	53	21,686	21,739
▼ src	0.2 %	53	21,686	21,739
▼ code	0.2 %	53	21,686	21,739
▼ c96_SmartElevator.java	94.6 %	53	3	56
▼ c96_SmartElevator	94.6 %	53	3	56
● main(String[])	100.0 %	15	0	15
● c96_SmartElevator(double)	100.0 %	9	0	9
● updateLoad(double)	100.0 %	29	0	29

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c96_SmartElevator.java	100.0 %	2	0	2
▼ c96_SmartElevator	100.0 %	2	0	2
● main(String[])		0	0	0
● c96_SmartElevator(double)		0	0	0
● updateLoad(double)	100.0 %	2	0	2

9. Unimproved V-method instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.2 %	53	21,686	21,739
▼ src	0.2 %	53	21,686	21,739
▼ code	0.2 %	53	21,686	21,739
▼ c96_SmartElevator.java	94.6 %	53	3	56
▼ c96_SmartElevator	94.6 %	53	3	56
● main(String[])	100.0 %	15	0	15
● c96_SmartElevator(double)	100.0 %	9	0	9
● updateLoad(double)	100.0 %	29	0	29